US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12397359
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Miyamoto; Gen

Metalworking portable cutting machine

Abstract

A metalworking portable cutting machine has a lid for an outlet of a dust collector box with improved operability. A metalworking portable cutting machine includes a cutting machine body including a blade, and a dust collector box to collect a chip resulting from cutting. The dust collector box includes an outlet to discharge the chip, and a lid to open or close the outlet. The lid includes a heat dissipater on a part of a surface of the lid.

Inventors: **Miyamoto; Gen (Anjo, JP)**

Applicant: **MAKITA CORPORATION** (Anjo, JP)

Family ID: **1000008779877**

Assignee: **MAKITA CORPORATION (Anjo, JP)**

Appl. No.: **18/131220**

Filed: **April 05, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230390839 A1	Dec. 07, 2023

Foreign Application Priority Data

JP	2022-091564	Jun. 06, 2022

Publication Classification

Int. Cl.: **B23D59/00** (20060101)

U.S. Cl.:

CPC **B23D59/006** (20130101);

Field of Classification Search

CPC: B23D (59/006)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5321890	12/1993	LaBlue	30/381	B23Q 11/0053
6557261	12/2002	Buser	30/390	B23D 59/006
7628102	12/2008	Kamiya	83/100	B23D 59/006
2005/0120845	12/2004	Kamiya	83/78	B23Q 11/127
2006/0169111	12/2005	Kozlowski	83/13	B23D 59/003
2010/0000095	12/2009	Matsumoto	30/124	B23D 59/006
2010/0089497	12/2009	Keenan	409/137	B23Q 11/0046
2015/0328796	12/2014	Okouchi	30/388	B25F 5/02
2016/0016270	12/2015	Batres	83/100	B01D 46/0002
2016/0176064	12/2015	Okouchi	30/374	B27B 9/00
2016/0368166	12/2015	Numata	N/A	B23D 45/16
2021/0362252	12/2020	Kimura	N/A	B23D 45/16
2023/0201934	12/2022	Funabiki	30/390	B23D 59/006
2023/0390839	12/2022	Miyamoto	N/A	B23D 59/006

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
563-031921	12/1987	JP	N/A
2561668	12/1997	JP	N/A
2019-198912	12/2018	JP	N/A

Primary Examiner: Michalski; Sean M

Attorney, Agent or Firm: Oliff PLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Japanese Patent Application No. 2022-091564, filed on Jun. 6, 2022, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a metalworking portable cutting machine to be held and operated by an operator with hands for cutting metal.

2. Description of the Background

(3) For example, a metalworking portable cutting machine including a tipped saw blade for metal, referred to as a metal cutter, includes a dust collector box for collecting sparks resulting from cutting and high-temperature chips left after the sparks cool. As described in Japanese Unexamined

Patent Application Publication No. 2019-198912, a dust collector box has a rear outlet through which collected chips are to be removed. The outlet can be open and closed with a lid. The user opens the lid to remove the chips in the dust collector box through the outlet. A tipped saw blade is a type of cutting blade having tips and a metal base prepared separately and then welded together.
BRIEF SUMMARY
(4) The lid may be heated to high temperature with the chips in the dust collector box. The lid is thus to be open or closed with caution. The lid is thus to be more easily operable. One or more aspects of the present disclosure are directed to a metalworking portable cutting machine including a lid for an outlet of a dust collector box with improved operability.
(5) A first aspect of the present invention provides a metalworking portable cutting machine, including: a cutting machine body including a blade; and a dust collector box to collect a chip resulting from cutting, the dust collector box including an outlet to discharge the chip, and a lid to open or close the outlet, the lid including a heat dissipater on a part of a surface of the lid.
(6) In this structure, the heat dissipater allows efficient dissipation of heat from the lid. The lid is less likely to be hot and is easily operable when being open and closed.
(7) A second aspect of the present invention provides a metalworking portable cutting machine, including: a cutting machine body including a blade; and a dust collector box to collect a chip resulting from cutting, the dust collector box including a box body, the box body including a first body comprising a first synthetic resin, the first body including a metal plate integral with an inner surface of the first body, and a second body comprising a second synthetic resin being more heat-resistant than the first synthetic resin.
(8) The use of metal in the first body alone reduces the weight of the dust collector box and allows the second body to improve the heat resistance of the dust collector box.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a right side view of a metalworking portable cutting machine according to an embodiment.
(2) FIG. **2** is a plan view of the metalworking portable cutting machine as viewed in the direction indicated by arrow II in FIG. **1**.
(3) FIG. **3** is a front view of the metalworking portable cutting machine as viewed in the direction indicated by arrow III in FIG. **1**.
(4) FIG. **4** is a rear view of the metalworking portable cutting machine as viewed in the direction indicated by arrow IV in FIG. **1**.
(5) FIG. **5** is a cross-sectional view taken along line V-V in FIG. **1**, as viewed in the direction indicated by arrows.
(6) FIG. **6** is a cross-sectional view taken along line VI-VI in FIG. **1**, as viewed in the direction indicated by arrows.
(7) FIG. **7** is a right side view of the metalworking portable cutting machine with its dust collector box detached.
(8) FIG. **8** is a perspective view of the dust collector box as viewed diagonally from the right front.
(9) FIG. **9** is a perspective view of the dust collector box as viewed diagonally from the right rear.
(10) FIG. **10** is a perspective view of the dust collector box as viewed diagonally from the right rear, with its lid open.
(11) FIG. **11** is a perspective view of a first body as viewed diagonally from the left rear, showing the inside of a box body.
(12) FIG. **12** is a perspective view of a second body as viewed diagonally from the right rear, showing the inside of the box body.

(13) FIG. **13** is a longitudinal sectional view of the dust collector box, with the lid closed.
(14) FIG. **14** is a rear view of the lid.
(15) FIG. **15** is a longitudinal sectional view of the lid taken along line XV-XV in FIG. **14**, as viewed in the direction indicated by arrows.

DETAILED DESCRIPTION

(16) As shown in FIGS. **1** to **4**, a metalworking portable cutting machine **1** according to an embodiment includes a tipped saw blade for metal and is also referred to as a metal cutter. The metalworking portable cutting machine **1** includes a base **10** and a cutting machine body **20**. The base **10** is placed in contact with the upper surface of a workpiece **2**. The cutting machine body **20** is supported on the upper surface of the base **10**. The base **10** is a rectangular flat plate with its lower surface in contact with the upper surface of the workpiece **2**. The user is behind the metalworking portable cutting machine **1** and moves the metalworking portable cutting machine **1** forward for cutting. The front herein refers to the cutting direction. The rear herein refers to the direction toward the user. The lateral direction is defined as viewed from the user.
(17) As shown in FIGS. **5** to **7**, the cutting machine body **20** includes an electric motor **21** as a driving source and a blade **30**. The blade **30** is a circular plate rotatable by the electric motor **21**. The electric motor **21** is, for example, a high-power brushless motor that is compact in a direction along its axis. Rotation from the electric motor **21** is output to an output shaft **23** through a reduction gear **22**. A gear housing **25** is connected to the right end of a motor housing **24**.
(18) A loop handle **26** is located above the joint between the motor housing **24** and the gear housing **25**. A switch lever **27** is located on the inner periphery of the handle **26**. The user turns on the switch lever **27** with the fingers holding the handle **26** to activate the electric motor **21**.
(19) A battery mount **28** is located at the rear of the handle **26**. The battery mount **28** receives a battery pack **11**. The battery pack **11** is, for example, a lithium-ion battery attachable in a slidable manner. The electric motor **21** is powered by the battery pack **11** mounted on the battery mount **28**. The battery pack **11** detached from the battery mount **28** may be charged with a separate charger to allow repeated use.
(20) As shown in FIGS. **5** to **7**, a stationary cover **35** formed from a transparent resin is connected to a flange **25***a* at the right end of the gear housing **25**. The stationary cover **35** is formed from polycarbonate and is connected to the gear housing **25** with two fastener screws **35***d*. The stationary cover **35** and the flange **25***a* on the gear housing **25** are connected together into a cover. The output shaft **23** protrudes into the stationary cover **35** to receive the blade **30**. The blade **30** is held between an outer flange **31** and an inner flange **32** in a direction along its thickness. In this state, the blade **30** is fastened to the output shaft **23** with a fastener screw **33**. The blade **30** is, for example, a circular saw blade for cutting metal materials, which is also referred to as a tipped saw blade.
(21) The blade **30** has its substantially upper half (portion above the upper surface of the base **10**) covered with the stationary cover **35** and the flange **25***a* on the gear housing **25**. The blade **30** cutting into the workpiece **2** produces chips in a cutting portion **3**. The chips are blown up by the rotating blade **30**. The stationary cover **35** includes a rectangular dust collector duct **35***b* at its front. The dust collector duct **35***b* is located above the cutting portion **3**. A metal plate **35***c*, which is a plate of steel, is located on the front inner peripheral surface of the dust collector duct **35***b*. The metal plate **35***c* protects the stationary cover **35** from high-temperature chips hitting the stationary cover **35**. The chips blown up from the cutting portion **3** are thus guided through the dust collector duct **35***b* and flow into a dust collector box **40**.
(22) An arrow **35***a* on the right side of the stationary cover **35** indicates the rotation direction of the blade **30**.
(23) The blade **30** has its substantially lower half (portion below the lower surface of the base **10**) covered with a movable cover **36**. The movable cover **36** is rotatably supported on the right side of the gear housing **25**. As the movable cover **36** rotates, the cutting edge of the blade **30** is exposed below the lower surface of the base **10**. The movable cover **36** is urged in the closing direction. As

the metalworking portable cutting machine **1** is moved for cutting, the movable cover **36** pressed against the workpiece **2** gradually opens against the urging force.

(24) The cutting machine body **20** is supported on the upper surface of the base **10** with a swing support **37**. The swing support **37** allows the cutting machine body **20** to swing vertically. The depth of a cut created by the blade **30** into the workpiece **2** is adjustable by changing the vertical swing position of the cutting machine body **20**.

(25) The dust collector box **40** is detachably attached to the right side of the stationary cover **35**. The dust collector box **40** collects chips flowing into it from inside the stationary cover **35**. FIGS. **8** to **10** show the dust collector box **40** detached from the right side of the stationary cover **35**. The dust collector box **40** includes a box body **41** formed from a synthetic resin. An arrow **30***a* on the right side of the box body **41** indicates the rotation direction of the blade **30** (counterclockwise in FIG. **1**).

(26) The box body **41** extends in the front-rear and vertical directions and is short in the lateral width direction (along the thickness of the blade **30**). The box body **41** includes right and left halves. More specifically, the box body **41** includes a first body **42** on the right and a second body **43** on the left. The first body **42** and the second body **43** face each other and are connected together into the box body **41**. The joint between the first body **42** and the second body **43** is denoted by character J in the drawings. As shown in FIG. **13**, the first body **42** and the second body **43** are connected together with three fastener screws **41***a* in their front, rear, and upper portions. The three fastener screws **41***a* are placed through the second body **43** and tightened. The fastener screws **41***a* are thus invisible on the first body **42**.

(27) FIG. **11** shows the inner surface of the first body **42** separate from the second body **43**. The first body **42** has three threaded holes **42***b* at its periphery for the fastener screws **41***a* to be tightened into. The first body **42** is formed from polycarbonate (first synthetic resin), which is a light-transmissive material. An aluminum heat-resistant plate **42***a* is connected to the inner surface of the first body **42** by press fitting. As shown in FIGS. **5** and **11**, the heat-resistant plate **42***a* covers an area of the inner surface of the first body **42** extending rearward from the center of the blade **30**. An area of the inner surface of the first body **42** extending frontward from the center of the blade **30** and a part of the intermediate area are covered with the heat-resistant plate **42***a* substantially in their lower half. The substantially upper half of these areas uncovered with the heat-resistant plate **42***a* and thus light-transmissive allows visual check of the volume of collected chips. In FIGS. **1** and **2**, this light-transmissive area of the first body **42** (light-transmissive portion **40***a*) for the visual check of the volume of chips inside is cross-hatched.

(28) As shown in FIGS. **6** and **11**, the inner surface of the first body **42** is covered with the heat-resistant plate **42***a* throughout its vertical range in its intermediate area and an area rearward from the center of the blade **30**. This improves the heat resistance of the first body **42** throughout its vertical range.

(29) FIG. **12** shows the inner surface of the second body **43** separate from the first body **42**. The second body **43** is formed from glass fiber-containing polyamide (second synthetic resin), which is a highly heat-resistant material. As shown in FIGS. **2**, **3**, **5**, and **6**, the second body **43** has a width W**2** greater than a width W**1** of the first body **42** in the direction along the thickness of the blade **30**. The greater width W**2** of the highly heat-resistant second body **43** further improves the heat resistance of the dust collector box **40**.

(30) As shown in FIG. **12**, the second body **43** has a dust collecting port **43***a* at the front on its inner surface. The chips blown up into the stationary cover **35** are carried with an airflow generated by the rotating blade **30**, guided through the dust collector duct **35***b*, and collected into the box body **41** through the dust collecting port **43***a*. The dust collecting port **43***a* corresponds to the inlet of the dust collector box **40**. The dust collecting port **43***a* is located above the dust collector duct **35***b*.

(31) As shown in FIG. **6**, the dust collector box **40** is fastened to the stationary cover **35** with a fastener screw **44**. As shown in FIGS. **6** and **7**, the fastener screw **44** is tightened into a threaded

hole **25***b* in the flange **25***a* on the right side of the gear housing **25**. As shown in FIGS. **1** and **8** to **10**, the fastener screw **44** is fitted with a large knob **45** at its right end. This improves the operability of the fastener screw **44** in tightening and loosening operations. The fastener screw **44** is loosened and removed from the threaded hole **25***b* to detach the dust collector box **40**. The knob **45** is received in a circular recess **40***b* on the right side of the box body **41**. This structure allows the dust collector box **40** to be compact in the lateral width direction without the knob **45** protruding.

(32) As shown in FIGS. **10** and **13**, the box body **41** has an outlet **46** for discharging chips and a vent **47** at its rear. The outlet **46** is a rectangular opening. The outlet **46** can be open and closed with a lid **50**. As shown in FIGS. **8** to **10**, the lid **50** is supported in a manner vertically pivotable about a support shaft **41***b* located at the rear of the box body **41**. The support shaft **41***b* protrudes both to the right and to the left.

(33) Similarly to the second body **43** in the box body **41**, the lid **50** is formed from glass fiber-containing polyamide. As shown in FIGS. **14** and **15**, the lid **50** has a pair of hinges, or a right hinge **51** and a left hinge **52**, and an arm **53**. The pair of right and left hinges **51** and **52** are connected to right and left portions of the support shaft **41***b*. The arm **53** extends from ends of the hinges **51** and **52** at an angle with the hinges **51** and **52**. The right and left hinges **51** and **52** have connecting holes **51***a* and **52***a*. The support shaft **41***b* is placed through each of the connecting holes **51***a* and **52***a* to support the lid **50** in a vertically rotatable manner.

(34) As shown in FIG. **14**, the right hinge **51** has a thickness d**1** greater than a thickness d**2** of the left hinge **52**. The right hinge **51** is thus harder than the left hinge **52**. This facilitates attachment of the lid **50** to the box body **41**. The harder right hinge **51** is connected to the support shaft **41***b* first. The less hard left hinge **52** can thus be easily connected to the support shaft **41***b* later. As shown in FIGS. **8** to **10**, the right hinge **51** has a recess **51***b* on its right side. This reduces the cost for molding.

(35) As shown in FIGS. **13** to **15**, the arm **53** has a recess **54**. The recess **54** is recessed toward the hinges **51** and **52** (frontward). The recess **54** includes a wall **54***a* and a bottom **54***b*. The wall **54***a* is looped and protrudes frontward. The wall **54***a* has its end covered with the bottom **54***b*. The recess **54** receives a heat dissipater **55**. The heat dissipater **55** enhances the heat dissipation of the lid **50**. This improves the operability of the lid **50** in opening and closing operations.

(36) The heat dissipater **55** includes multiple (four in the present embodiment) ridges **55***a*. The four ridges **55***a* are received in the recess **54**. Each ridge **55***a* extends laterally. The four ridges **55***a* are regularly spaced and extend parallel to one another. The four ridges **55***a* extend from the bottom **54***b* of the recess **54** perpendicularly to the direction of rotation of the lid **50**, or parallel to the tangent to an arc centered on the support shaft **41***b*. The ends of the four ridges are connected to the wall **54***a* of the recess **54**. The wall **54***a* connects the ends of the four ridges **55***a* together, improving the rigidity of each ridge **55***a* as a support. The four ridges **55***a* also improve the rigidity of the recess **54**.

(37) The four ridges **55***a* and the recess **54** increase the external surface area of the lid **50** to enhance the heat dissipation of the lid **50**. As shown in FIG. **13**, the recess **54** is located in the outlet **46** when the lid **50** is closed. The dust collector box **40** can thus have a compact rear portion when the lid **50** is closed. The recess **54** and the heat dissipater **55** located in the outlet **46** also facilitate the discharge of heat from the outlet **46** and its surroundings.

(38) The lid **50** includes an engagement tab **56**. The engagement tab **56** extends from an end of the arm **53** at an angle toward the hinges **51** and **52**. The engagement tab **56** has a rectangular engagement hole **57**. The engagement hole **57** has its lower edge elastically engaged with an engagement portion **49** on the rear surface of the dust collector box **40** when the lid **50** is closed. This elastic engagement between the lower edge of the engagement hole **57** and the engagement portion **49** holds the lid **50** being closed at the position. The engagement portion **49** has its rear surface sloping frontward to the lower end. This allows easier operation in opening and closing the lid **50** (the operability of the lid **50** in opening and closing operations) for causing the lower edge of

the engagement hole **57** to be engaged with or disengaged from the engagement portion **49**.
(39) As shown in FIGS. **9**, **10**, and **13**, the dust collector box **40** has the vent **47** in its rear surface. The vent **47** is covered with a metal air-vent plate **48** commonly referred to as perforated metal. The air-vent plate **48** has multiple small air vents **48***a*. The air-vent plate **48** improves the heat resistance of the vent **47** and an airflow through the vent **47**. As shown in FIGS. **9** and **13**, the vent **47** faces an upper portion of the engagement hole **57** when the lid **50** is closed. The vent **47** is not blocked when the lid **50** is closed. This allows smooth airflow in the box body **41** and efficient discharge of heat from the box body **41** through the vent **47** when the lid **50** is closed.
(40) A projection **58** is located at the lower end of the engagement tab **56**. The projection **58** extends away from the box body **41**. The projection **58** is pulled away from the box body **41** to detach the engagement portion **49** from the lower edge of the engagement hole **57**. The projection **58** is then pulled upward to open the lid **50**. To close the lid **50**, the projection **58** is pulled downward toward the box body **41**. This engages the lower edge of the engagement hole **57** with the engagement portion **49**. The projection **58** improves the operability of the lid in opening and closing operations.
(41) The metalworking portable cutting machine **1** according to the present embodiment includes the dust collector box **40** with the outlet **46** and the lid **50** to open or close the outlet **46**. The lid **50** includes the heat dissipater **55**. This allows efficient dissipation of heat from the lid through the heat dissipater **55** and thus improves the operability of the lid **50** in opening and closing operations.
(42) The dust collector box **40** in the embodiment includes the box body **41** having right and left halves, or the first body **42** and the second body **43**. The first body **42** is formed from polycarbonate (first synthetic resin) with the aluminum heat-resistant plate **42***a* integral with its inner surface. The second body **43** is formed from glass fiber-containing polyamide (second synthetic resin), which is a material more heat-resistant than polycarbonate. The use of the metal component in the first body **42** alone reduces the weight of the dust collector box **40** and allows the second body **43** to improve the heat resistance of the dust collector box **40**.
(43) The heat dissipater **55** in the embodiment includes the multiple ridges **55***a*. This increases the surface area of the lid **50** to allow efficient dissipation of heat from the lid **50**.
(44) The multiple ridges **55***a* in the embodiment are parallel to one another. This creates the channels for outside air to flow between the ridges **55***a* to enhance the heat dissipation of the lid **50**.
(45) The lid **50** in the embodiment is rotatably connected to the box body **41** of the dust collector box **40**. The multiple ridges **55***a* extend perpendicularly to the direction of rotation of the lid **50**. The lid **50** includes the wall **54***a* connecting the ends of the multiple ridges **55***a* together. This improves the rigidity of the multiple ridges **55***a* as a support.
(46) The lid **50** in the embodiment includes the hinges **51** and **52** rotatably connected to the box body **41** of the dust collector box **40**. The arm **53** extends from the ends of the hinges **51** and **52** at an angle with the hinges **51** and **52**. The arm **53** has the recess **54** recessed toward the hinges **51** and **52**. The arm **53** with this structure is less likely to deform and thus improves the rigidity of the lid **50** without increasing its size. The arm **53** having the recess **54** with its increased surface area also enhances heat dissipation.
(47) The recess **54** in the embodiment is located in the outlet **46** when the lid **50** is closed. The dust collector box **40** can have a compact rear portion when the lid **50** is closed.
(48) The recess **54** in the embodiment includes the wall **54***a* that is looped and extends from the arm **53** toward the hinges **51** and **52** and the bottom **54***b* covering the end of the wall **54***a*. The heat dissipater **55** includes the multiple ridges **55***a* protruding from the bottom **54***b* toward the arm **53**. The heat dissipater **55** includes the multiple ridges **55***a* received in the recess **54** and thus is compact.
(49) The lid **50** in the embodiment includes the engagement tab **56** extending from the end of the arm **53** at an angle toward the hinges **51** and **52**. The engagement tab **56** has the engagement hole **57** engageable with the engagement portion **49** of the dust collector box **40**. This structure holds the

lid **50** being closed at the position. The hinges **51** and **52**, the arm **53**, and the engagement tab **56** are arranged in a curve along the rear surface of the box body **41** to improve the rigidity of the lid **50**. This structure improves the operability of the lid **50** in opening and closing operations.
(50) The dust collector box **40** in the embodiment has the vent **47** to allow entry of outside air into the dust collector box **40**. The vent **47** is covered with the metal air-vent plate **48** with multiple air vents **48***a*. The engagement hole **57** in the lid **50** faces the vent **47** (air vents **48***a*). This improves airflow through the vent **47** (air vents **48***a*) when the lid **50** is closed.
(51) The lid **50** in the embodiment includes the projection **58** extending from the end of the engagement tab **56** away from the box body **41**. This structure improves the operability of the lid **50** in opening and closing operations.
(52) The polycarbonate (first synthetic resin) in the first body **42** in the embodiment is light-transmissive. This allows visual check of the volume of collected chips through the light-transmissive portion **40***a*.
(53) The box body **41** in the embodiment includes the first body **42** in a first direction along the thickness of the blade **30** and the second body **43** in a second direction along the thickness of the blade **30**. The second body **43** has the width W**2** along the thickness greater than the width W**1** of the first body **42** along the thickness. This allows the dust collector box to have high heat resistance.
(54) The second body **43** in the dust collector box **40** in the embodiment is formed from a material containing polyamide as the second synthetic resin and also containing glass fiber. This allows the second body **43** to have high heat resistance.
(55) The embodiment described above may be modified variously. For example, although the four ridges **55***a* on the heat dissipater **55** extend laterally in the above embodiment, the ridges **55***a* may extend vertically. In some embodiments, the heat dissipater **55** may include multiple ridges arranged in a lattice pattern.
(56) Although the multiple ridges **55***a* are received in the recess **54** with their tops lower than the surface of the arm **53** in the above embodiment, the recess **54** may be eliminated, or the multiple ridges may protrude from a part of the surface of the arm **53**.
(57) Although the thicknesses d**1** and d**2** of the right and left hinges **51** and **52** of the lid **50** are different in the above embodiment, the hinges **51** and **52** may have the same thickness. The recess **51***b*, which is a lightening portion, may be eliminated.
(58) Although the lid **50** vertically pivots to be open and closed in the above embodiment, the lid may be vertically slidable to be open and closed and may include the heat dissipater **55** in the above embodiment.
(59) The second body **43** may be formed from a highly heat-resistant resin in place of glass fiber-containing polyamide.
(60) Similarly to the second body **43**, the lid **50** may be formed from glass fiber-containing polyamide or any other highly heat-resistant resin.
(61) The heat-resistant plate **42***a* in the first body **42** may be a heat-resistant plate of iron rather than aluminum.
(62) The first body **42** in the box body **41** may also be formed from glass fiber-containing polyamide, similarly to the second body **43**, except at a window for the visual check of the volume of collected chips. In this case, the heat-resistant plate **42***a* may be eliminated. The window for the visual check of the volume of collected chips may be a separate transparent plate formed from a light-transmissive polycarbonate and fitted into the window. The box body may be a one-piece structure, for example produced by blow molding, rather than having right and left halves.
(63) The metalworking portable cutting machine may include a diamond wheel or a grinding disc rather than being a metal cutter including a tipped saw blade for metal.
REFERENCE SIGNS LIST
(64) **1** metalworking portable cutting machine (metal cutter) **2** workpiece **3** cutting portion **10** base

11 battery pack **20** cutting machine body **21** electric motor (brushless motor) **22** reduction gear **23** output shaft **24** motor housing **25** gear housing **25***a* flange **25***b* threaded hole **26** handle **27** switch lever **28** battery mount **30** blade **30***a* arrow (rotation direction of blade **30**) **31** outer flange **32** inner flange **33** fastener screw **35** stationary cover **35***a* arrow (rotation direction of blade **30**) **35***b* dust collector duct **35***c* metal plate **35***d* fastener screw **36** movable cover **37** swing support **40** dust collector box **40***a* light-transmissive portion **40***b* recess **41** box body **41***a* fastener screw **41***b* support shaft **42** first body **42***a* heat-resistant plate **42***b* threaded hole W**1** width of first body **42** **43** second body **43***a* dust collecting port W**2** width of second body **43** **44** fastener screw **45** knob **46** outlet **47** vent **48** air-vent plate **48***a* air vent **49** engagement portion **50** lid **51** hinge (right) **51***a* connecting hole **51***b* recess (lightening portion) d**1** thickness of hinge **51** **52** hinge (left) **52***a* connecting hole d**2** thickness of hinge **52** **53** arm **54** recess **54***a* wall **54***b* bottom **55** heat dissipater **55***a* ridge **56** engagement tab **57** engagement hole **58** projection

Claims

1. A metalworking portable cutting machine, comprising: a cutting machine body including a blade; and a dust collector box to collect a chip resulting from cutting, the dust collector box including a box body, the box body including a first body comprising a first synthetic resin, the first body including a metal plate integral with an inner surface of the first body, and a second body comprising a second synthetic resin being more heat-resistant than the first synthetic resin.
2. The metalworking portable cutting machine according to claim 1, wherein the dust collector box includes an outlet to discharge the chip, and a lid to open or close the outlet, and the lid includes a hinge rotatably connected to the box body, an arm extending from an end of the hinge at an angle with the hinge, and a recess recessed from the arm toward the hinge.
3. The metalworking portable cutting machine according to claim 2, wherein the recess is located in the outlet with the lid being closed.
4. The metalworking portable cutting machine according to claim 2, wherein the recess includes a wall being looped and extending from the arm toward the hinge, and a bottom covering an end of the wall.
5. The metalworking portable cutting machine according to claim 2, wherein the lid includes an engagement tab extending from an end of the arm at an angle toward the hinge, and the engagement tab has an engagement hole engageable with an engagement portion of the dust collector box.
6. The metalworking portable cutting machine according to claim 5, wherein the dust collector box includes a metal air-vent plate having a plurality of air vents, and the engagement hole faces the metal air-vent plate.
7. The metalworking portable cutting machine according to claim 5, wherein the lid includes a projection extending from an end of the engagement tab away from the box body.
8. The metalworking portable cutting machine according to claim 1, wherein the first synthetic resin in the first body is light-transmissive.
9. The metalworking portable cutting machine according to claim 8, wherein the first body is located in a first direction along a thickness of the blade, the second body is located in a second direction along the thickness of the blade, and the second body has a width along the thickness greater than a width of the first body along the thickness.
10. The metalworking portable cutting machine according to claim 1, wherein the first body is located in a first direction along a thickness of the blade, the second body is located in a second direction along the thickness of the blade, and the second body has a width along the thickness greater than a width of the first body along the thickness.
11. The metalworking portable cutting machine according to claim 1, wherein the second body comprises a material containing polyamide as the second synthetic resin and containing glass fiber.

12. The metalworking portable cutting machine according to claim 8, wherein the second body comprises a material containing polyamide as the second synthetic resin and containing glass fiber.